

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)			no more effervescence		1		1		1
	(b)			award (1) each for any two improvements and (1) each for linked explanations use larger beaker / conical flask (1) prevents acid spray from escaping (1) rinse beaker and add washings to flask (1) ensures no acid left in beaker (1) use more acid / less chalk (1) gives larger titres therefore smaller percentage error (1) crush the chalk / heat the solution / stir the solution (1) speeds up the reaction (1)			4	4		4
	(c)			$n(\text{NaOH}) = 1.64 \times 10^{-3} \text{ mol}$ (1) excess $n(\text{HCl}) = 1.64 \times 10^{-3} \text{ mol}$ $n(\text{HCl}) \text{ reacted} = 0.05 - 1.64 \times 10^{-3} = 0.0484 \text{ mol}$ (1) $n(\text{CaCO}_3) = 0.0242 \text{ mol}$ mass $\text{CaCO}_3 = 2.42 \text{ g}$ (1) percentage $\text{CaCO}_3 = 95.3 \%$ (1)		4		4	3	4
	(d)			not valid as first titre should be disregarded / is rough value			1	1		1

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					AO1	AO2	AO3	Total	Maths	Prac
	(e)			value of titre would increase (1) fewer moles HCl react with limestone therefore more moles NaOH needed to react with excess (1)			2	2		2
				Question 11 total	0	5	7	12	3	12